AI Quant Research Lab

Comprehensive Research Report

Research asset: Apple Inc. (AAPL) Research period: 2021–2025 Report type: Historical quantitative research and strategy evaluation

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1. Executive Summary

This project investigates whether systematic trading strategies and machine-learning models can identify repeatable opportunities in historical equity-market data.

The research emphasizes out-of-sample evaluation, transaction costs, statistical testing, parameter sensitivity and explicit rejection of strategies that fail to demonstrate convincing evidence.

The experiment registry currently contains 3 experiments, of which 3 are rejected.

The findings do not establish a reliable benchmark-beating trading edge. Instead, the experiments demonstrate the importance of realistic execution assumptions, statistical validation and disciplined research methodology.

2. Research Objective

The objective is to develop a research framework that attempts to disprove its own trading hypotheses rather than simply identify historical strategies with attractive returns.

The framework evaluates:

• Signal generation and execution timing.

• Historical performance against buy-and-hold.

• Transaction-cost sensitivity.

• Out-of-sample performance.

• Statistical significance.

• Parameter sensitivity.

• Dependence on unusually strong trading days.

• Predictive performance of machine-learning models.

Research principle

A strategy is not accepted merely because it produces positive historical returns. It must demonstrate credible performance after costs, against an appropriate benchmark, and under out-of-sample evaluation.

3. Moving-Average Strategy

The moving-average strategy uses a short-term simple moving average (SMA) and a long-term SMA.

A long position is held when the short-term average is above the long-term average. The previous session's signal determines the current session's position.

The historical backtest uses adjusted closing prices and assumes a 0.10% transaction cost per position change.

Earlier parameter sensitivity experiment

An earlier experiment evaluated four parameter combinations: 10/30, 20/50, 30/100 and 50/150.

All configurations use a common evaluation start determined by the first valid 150-session moving average. The notebook's research period is January 2021 to December 2025.

Short MA	Long MA	Strategy Return	Strategy CAGR	Strategy Sharpe	Strategy Max Drawdown	Benchmark Return	Number of Position Changes
10	30	22.30%	4.69%	0.3408	-27.95%	89.30%	41
20	50	29.88%	6.14%	0.4157	-29.80%	89.30%	26
30	100	33.24%	6.76%	0.4372	-31.41%	89.30%	12
50	150	11.79%	2.57%	0.2275	-28.40%	89.30%	11

The earlier parameter experiment is a historical sensitivity analysis, not an independent out-of-sample test.

Later six-configuration analysis

A separate analysis evaluated six moving-average configurations:

Short_Window	Long_Window	Total Return	CAGR	Annualized Volatility	Sharpe Ratio (rf=0)	Maximum Drawdown	Buy_Hold_Total_Return	Buy_Hold_Sharpe	Positive_Return	Beats_Buy_Hold_Return	Beats_Buy_Hold_Sharpe
10	30	28.77%	5.21%	17.94%	0.3723	-27.95%	115.81%	0.6932	True	False	False
15	40	38.51%	6.76%	17.28%	0.4648	-32.34%	115.81%	0.6932	True	False	False
20	50	33.15%	5.92%	17.83%	0.4113	-29.80%	115.81%	0.6932	True	False	False
25	60	67.38%	10.90%	17.77%	0.6709	-24.22%	115.81%	0.6932	True	False	False
30	100	44.69%	7.70%	18.23%	0.4979	-31.41%	115.81%	0.6932	True	False	False
50	150	11.79%	2.26%	18.52%	0.2136	-28.40%	115.81%	0.6932	True	False	False

The two parameter experiments should be interpreted separately. Their results should not be combined because their evaluation procedures and source data may differ.

Interpretation

The moving-average experiments did not establish consistent outperformance over buy-and-hold. Parameter sensitivity also demonstrates that historical performance depends on the selected moving-average windows.

The six-configuration analysis identified a stronger historical configuration, but it still did not outperform buy-and-hold on total return or Sharpe ratio.

4. Mean-Reversion Strategy — Close Execution

The close-execution mean-reversion experiment evaluates whether prices sufficiently below a rolling average subsequently recover.

Parameters were selected using the 2021–2023 training period and evaluated on 2024–2025 data.

The selected configuration was a 20-session window with a -5% entry threshold.

Out-of-sample performance

Metric	Mean Reversion	Buy and Hold
Total Return	8.72%	42.54%

Metric	Mean Reversion	Buy and Hold
CAGR	4.29%	19.47%
Annualized Volatility	15.18%	27.99%
Sharpe Ratio (rf=0)	0.3495	0.7742
Maximum Drawdown	-18.96%	-33.36%

HAC significance test

Observations	Mean_Daily_Excess_Return	Ordinary_T_Statistic	Ordinary_P_Value	HAC_Lags	HAC_Standard_Error	HAC_T_Statistic	HAC_P_Value	HAC_CI_95_Lower	HAC_CI_95_Upper
502	-0.06%	-0.9808	0.3272	5	0.07%	-0.9266	0.3546	-0.20%	0.07%

Transaction-cost sensitivity

Transaction_Cost_Per_Turnover	Strategy_Total_Return	Strategy_CAGR	Strategy_Sharpe	Strategy_Max_Drawdown	Buy_Hold_Total_Return	Buy_Hold_Sharpe
0.00%	10.93%	5.34%	0.4162	-18.62%	42.54%	0.7742
0.05%	9.82%	4.82%	0.3828	-18.79%	42.54%	0.7742
0.10%	8.72%	4.29%	0.3495	-18.96%	42.54%	0.7742
0.25%	5.50%	2.72%	0.2496	-19.45%	42.54%	0.7742
0.50%	0.32%	0.16%	0.0841	-20.48%	42.54%	0.7742

The HAC p-value was approximately 0.3546. The experiment therefore did not demonstrate statistically significant mean daily excess returns at the 5% level.

The strategy also underperformed buy-and-hold on total return and Sharpe ratio.

Decision: REJECT.

5. Mean-Reversion Strategy — Open Execution

A separate implementation evaluates next-session execution using opening prices.

A signal observed at the previous close determines the position held during the next open-to-open return interval.

Parameters were selected using 2021–2023 training data and evaluated out of sample over 2024–2025.

The selected configuration was a 20-session window with a -5% entry threshold.

Out-of-sample performance

Metric	Mean Reversion (Open Execution)	Buy and Hold (Open-to-Open)
Total Return	11.50%	47.28%
CAGR	5.63%	21.50%
Annualized Volatility	17.21%	29.90%
Sharpe Ratio (rf=0)	0.4018	0.7998
Maximum Drawdown	-13.44%	-33.33%

The open-execution experiment produced a total return of approximately 11.50%, compared with 47.28% for its open-to-open buy-and-hold benchmark.

Its HAC p-value, as recorded in the experiment registry, was approximately 0.3404. This does not provide statistically significant evidence of positive mean daily excess returns at the 5% level.

Lower volatility and drawdown do not compensate for the substantial underperformance against the benchmark.

Decision: REJECT.

6. Static Machine-Learning Experiment

A logistic-regression classifier was evaluated using historical price-derived features.

The experiment investigated whether the model could predict the direction of subsequent returns.

Prediction metrics

Model	Accuracy	Balanced Accuracy	Precision	Recall	ROC AUC
Logistic Regression	0.4731	0.4583	0.5136	0.6227	0.4661

Model	Accuracy	Balanced Accuracy	Precision	Recall	ROC AUC
Training Majority Class	0.5449	0.5	0.5449	1	Not available
Always Predict Up	0.5449	0.5	0.5449	1	Not available

Strategy performance

Metric	ML Strategy	Buy and Hold
Total Return	-0.82%	47.83%
CAGR	-0.41%	21.73%
Annualized Volatility	23.64%	27.89%
Sharpe Ratio (0% risk-free)	0.0990	0.8429
Maximum Drawdown	-31.18%	-33.36%

The model's ROC AUC was approximately 0.4661, below the 0.50 level associated with random ranking.

The resulting strategy also underperformed buy-and-hold.

These results do not support the use of this model as a profitable trading signal in its tested configuration.

7. Walk-Forward Validation

Walk-forward validation evaluates a model sequentially using historical information available before each prediction.

This is a more realistic evaluation design than fitting a model once and evaluating it on a fixed holdout period.

Prediction metrics

Metric	Value
Accuracy	0.513
Balanced Accuracy	0.4848
ROC AUC	0.4469
Brier Score	0.2542

Metric	Value
Majority-Class Baseline Accuracy	0.5449
Historical-Rate Baseline Brier Score	0.2485

Strategy performance

Metric	Walk-Forward Strategy	Buy and Hold
Total Return	13.00%	47.83%
CAGR	6.34%	21.73%
Annualized Volatility	25.38%	27.89%
Sharpe Ratio	0.3676	0.8429
Max Drawdown	-31.73%	-33.36%

The walk-forward model did not establish predictive performance above its simple baselines. Its trading strategy also underperformed buy-and-hold.

Walk-forward evaluation improves the testing design, but does not by itself establish that a model is free from data leakage or implementation bias. Those properties depend on the complete feature-generation and execution pipeline.

8. Strategy Dependency and Market-Regime Diagnostics

Earlier moving-average best-day dependency

The earlier SMA experiment replaced its five strongest daily strategy returns with zero and recalculated performance.

This is an observation-deletion sensitivity test, not an executable alternative strategy.

Metric	Original Strategy	Without Best 5 Days
Total Return	26.54%	-0.68%
CAGR	12.54%	-0.34%
Annualized Volatility	18.52%	16.67%
Sharpe Ratio (0% risk-free)	0.7304	0.0631

Metric	Original Strategy	Without Best 5 Days
Maximum Drawdown	-25.03%	-25.03%

Mean-reversion best-day dependency

A separate diagnostic examined the effect of removing the strongest daily returns from the close-execution mean-reversion strategy and its benchmark.

Series	Test	Days_Removed	Total_Return	Return_Change	Observations_Remaining
Mean Reversion Strategy	Baseline	0	8.72%	0.00%	502
Mean Reversion Strategy	Remove top 1 days	1	-5.73%	-14.45%	501
Mean Reversion Strategy	Remove top 3 days	3	-12.30%	-21.03%	499
Mean Reversion Strategy	Remove top 5 days	5	-17.02%	-25.75%	497
Mean Reversion Strategy	Remove top 10 days	10	-25.10%	-33.83%	492
Buy and Hold	Baseline	0	42.54%	0.00%	502
Buy and Hold	Remove top 1 days	1	23.59%	-18.95%	501
Buy and Hold	Remove top 3 days	3	8.38%	-34.16%	499
Buy and Hold	Remove top 5 days	5	-2.69%	-45.23%	497
Buy and Hold	Remove top 10 days	10	-20.69%	-63.23%	492

The two dependency analyses use different procedures and should not be directly compared.

Extreme-day analysis

The following table summarizes five of the strongest and five of the weakest recorded daily observations.

Extreme_Group	Date	Position	Asset_Return_Pct	Gross_Return_Pct	Transaction_Cost_Pct	Net_Return_Pct
Best 10	2025-04-09	1	15.33%	15.33%	0.00%	15.33%
Best 10	2025-04-11	1	4.06%	4.06%	0.00%	4.06%
Best 10	2025-04-22	1	3.41%	3.41%	0.10%	3.31%
Best 10	2025-01-27	1	3.18%	3.18%	0.00%	3.18%
Best 10	2025-05-27	1	2.53%	2.53%	0.10%	2.43%

Extreme_Group	Date	Position	Asset_Return_Pct	Gross_Return_Pct	Transaction_Cost_Pct	Net_Return_Pct
Worst 10	2025-03-13	1	-3.36%	-3.36%	0.00%	-3.36%
Worst 10	2025-04-07	1	-3.67%	-3.67%	0.00%	-3.67%
Worst 10	2025-04-10	1	-4.24%	-4.24%	0.00%	-4.24%
Worst 10	2025-04-08	1	-4.98%	-4.98%	0.00%	-4.98%
Worst 10	2025-04-04	1	-7.29%	-7.29%	0.10%	-7.39%

April 2025 market-regime observations

The following table selects the eight April 2025 sessions with the largest absolute underlying asset moves. It includes the strategy's signal, position, transaction cost and net return where those fields are available.

These are descriptive observations. They do not establish causal explanations for strategy performance.

Date	Asset_Return_Pct	Signal	Position	Transaction_Cost_Pct	Net_Return_Pct	Strategy_Equity	Buy_Hold_Equity
2025-04-03	-9.25%	1	0	0.00%	-0.00%	1.0229	1.0791
2025-04-04	-7.29%	1	1	0.10%	-7.39%	0.9473	1.0004
2025-04-07	-3.67%	1	1	0.00%	-3.67%	0.9125	0.9637
2025-04-08	-4.98%	1	1	0.00%	-4.98%	0.867	0.9157
2025-04-09	15.33%	1	1	0.00%	15.33%	1	1.056
2025-04-10	-4.24%	1	1	0.00%	-4.24%	0.9576	1.0113
2025-04-11	4.06%	1	1	0.00%	4.06%	0.9964	1.0523
2025-04-16	-3.89%	1	0	0.00%	-0.00%	1.0174	1.0317

9. Experiment Registry

The registry records formal strategy experiments and their decisions.

Experiment_ID	Strategy	Evaluation_Period	Strategy_Total_Return	Buy_Hold_Total_Return	Strategy_Sharpe	Buy_Hold_Sharpe	HAC_P_Value	Decision
AAPL_MR_OOS_2024_2025	Mean Reversion	2024-2025	8.72%	42.54%	0.3495	0.7742	0.3546	REJECT

Experiment_ID	Strategy	Evaluation_Period	Strategy_Total_Return	Buy_Hold_Total_Return	Strategy_Sharpe	Buy_Hold_Sharpe	HAC_P_Value	Decision
AAPL_SMA_BASELINE	Moving Average	2021-2025	33.15%	115.81%	0.4113	0.6932	Not available	REJECT
AAPL_MR_OPEN_OOS_2024_2025	Mean Reversion (Open Execution)	2024-2025	11.50%	47.28%	0.4018	0.7998	0.3404	REJECT

The registry currently contains 3 experiments, of which 3 are rejected.

The moving-average registry entry does not have a verified HAC significance result. Its unavailable p-value is not interpreted as evidence of statistical significance.

The close-execution and open-execution mean-reversion entries have recorded HAC p-values, neither of which indicates statistical significance at the 5% level.

10. Limitations

The research has several important limitations:

1. The primary experiments use a single equity asset, AAPL.

2. Historical performance does not guarantee future performance.

3. Transaction-cost assumptions simplify real-world execution.

4. Market impact, bid-ask spreads, liquidity constraints and partial fills are not fully modeled.

5. Some experiments use different execution conventions and evaluation periods, so their returns are not directly comparable.

6. Parameter sensitivity does not eliminate data-snooping risk.

7. Machine-learning features are primarily derived from historical price data.

8. The experiment registry contains a moving-average record without a verified HAC significance result.

9. Some older diagnostic outputs were generated independently in notebooks and have not been fully integrated into the formal experiment registry.

10. The framework is a research prototype, not a production trading system.

11. Future Research

Potential extensions include:

- Testing across multiple assets and market regimes.
- Adding point-in-time fundamental and alternative data.
- Modeling spreads, slippage and market impact.
- Applying purged and embargoed cross-validation where appropriate.
- Evaluating multiple-testing and data-snooping adjustments.
- Testing portfolio-level risk and exposure constraints.
- Improving feature engineering and model calibration.
- Comparing against stronger statistical and economic baselines.
- Requiring independent validation before accepting a strategy.

12. Final Conclusion

The experiments did not identify a strategy that convincingly outperformed buy-and-hold after the tested transaction costs and evaluation procedures.

The machine-learning experiments also failed to demonstrate reliable predictive superiority over simple baselines.

The main contribution of this project is its research framework: strategies are tested under explicit assumptions, evaluated against benchmarks, examined for statistical support and rejected when the evidence is insufficient.

The results should be interpreted as historical research findings, not investment advice or evidence of a deployable trading edge.